Telemetry Solutions in Disaggregated Optical Networks: an Experimental View

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Abstract: Effective telemetry streaming for disaggregated whitebox is discussed, focusing on implementation issues. A novel gRPC peer-to-peer telemetry for next-generation AI-equipped xPonders is proposed and evaluated in a disaggregated testbed exploiting OpenConfig and OpenROADM models.

1. Introduction

In the last years, the increasing interest on detailed optical performance monitoring has pushed researchers to define innovative methods to enable autonomic optical networking driven by accurate network awareness [1]. One of the most impacting methods is optical network telemetry. Recently, three main drivers have been playing a crucial role for the development of telemetry services. First, the introduction of network disaggregation at the optical layer, providing a common model to access, configure and monitor white boxes of multiple vendors, thus allowing a unified method to retrieve information from optical nodes, including xPonders (transponders or muxponders) and, recently, pluggable optical devices in an independent fashion [2]. The advantage to directly access all such devices compared to traditional closed, vendor- locked, monitoring systems, opens the way to novel capillary and distributed telemetry architectures, allowing to monitor many optical parameters, including, besides channel and aggregate optical power levels, those made available from the digital signal processing units of optical coherent receivers (e.g., Optical Signal to Noise Ratio -OSNR and pre Forwarding Error Correction Bit Error Rate – pre-FEC BER). Second, the availability of efficient open-source tools: lightweight streaming data protocols (e.g. gRPC), collector systems with time series database (e.g. Prometheus), and visualization tools (e.g. Grafana). Third, the advent of effective Artificial Intelligence (AI) platforms and algorithms (machine learning, neural networks) enabling fast big data analytics.

So far, centralized or hierarchical architectures have been considered for optical monitoring and telemetry, where a central collector receives streamed parameters from a subset/all nodes and feeds centralized workflows at the SDN Controller for network adaptation/optimization [3]. Such architectures have the capability to provide unprecedent network awareness. However, they are not free from limitations. Indeed, it is necessary to find a tradeoff between frequency of updates, capability to promptly react upon detrimental network events, and scalability issues. In addition, a central SDN Controller may not be fully aware of specific adaptations to enforce when proprietary/advanced transmission techniques are adopted (e.g., which FEC or constellation shaping type provides the best performance according to monitored parameters).

In this paper, we envision the most considered disaggregated telemetry architectures, protocols and implementations and we discuss existing open points from an experimental point of view. In particular, we highlight the possible drawbacks of centralized telemetry solutions, leading to scalability issues at the controller and possible awareness limitations. To address such issues, we propose a novel telemetry scheme suitable for disaggregated optical networking to bring network awareness locally at the xPonder pluggable optical card level. The proposed Peer-to-Peer Telemetry considers the collection at the source card of selected streaming data produced by a set of devices of interest (e.g., those crossed by the originated lightpath) and elaborated, at the card level, thanks to an internal lightweight AI module. The scheme is particularly suitable to identify and localize sporadic optical signal deviations, filter shifts or amplifier gain degradation and react promptly at the card level with automatic power or frequency readjustments, without involving the central SDN controller and within its configuration margins.

2. Telemetry and Disaggregation: Architectures, Implementations and Open Issues

Telemetry services, defined as augmented real-time monitoring streaming services, are designed on the degree of optical network disaggregation. In full disaggregation, telemetry is enabled between the SDN controller, the whitebox controllers and the controlled device agents, allowing hierarchical architectures. For example, a disaggregated node (e.g., ROADM) may be delegated to run local telemetry to controlled device agents and report the results to the control/monitoring plane. Such architecture is more complex but scalable since it allows a layered telemetry processing and correlation mechanism. In partial disaggregation, such layers are not feasible, and the SDN controller is typically the coordinator of telemetry subscriptions at the Optical Line System (OLS) and the xPonders.

Fig. 1 shows a telemetry service for a disaggregated network. Telemetry subscriptions are performed by the control plane (i.e., the SDN controller) to live monitor a set of devices crossed by a lightpath. In the figure, lightpath L_a is monitored through telemetry streams generated at L_a xPonder (e.g., OSNR, P_{in}, BER) and at intermediate ROADM2 (e.g., P_{in} and P_{out}). The subscription set may be enlarged, e.g. including xPonder of a co-routed adjacent L_b that may interfere with L_a. Streams are collected by central collectors (step 1), processed by AI-based Analytics Handler and, in case of deviations, a feedback is provided to the controller (step 2) in order to react properly (step 3) with recovery procedures (e.g., elastic operation, re-routing), also including fine tuning (e.g., launch power, signal adaptations, central frequency offset).

Scalability issues in terms of number of streaming sessions and rates may be relaxed thanks to selected protocols (e.g., IPFix, Thrift and gRPC), encompassing implicit headers, data bundling and compression, that require a limited additional complexity at the telemetry agent. Typically, if compared with traditional 15-minutes averaged statistics, telemetry with sampling period in the order of 1sec is enough to monitor effectively malfunctioning, soft failures, even though a number of open ROADMs platforms support up to 50ms telemetry update interval [4]. This means that, with respect to packet-switched networks running in-band telemetry where the service rate scales with data plane traffic rate, in optical telemetry a large amount of streaming sessions may be easily handled even by a central collector. Telemetry YANG models (i.e., OpenConfig, OpenROADM) are mature, including many service options and timers (i.e., streaming pace, collectors, groups, type of subscription, dial-in and dial-out option, heartbeat, encoding). Control-plane-embedded telemetry resorts to YANG/NETCONF Push notifications [5], while the most considered management plane -embedded telemetry, running in parallel with control plane, resorts to gRPC, including gNMI [6].

Telemetry scalability does not guarantee, however, that the centralized feedback-based awareness framework including monitors, Analytics Handler and controller, is scalable and effective. Indeed, a set of soft failures may be recovered conveniently at a lower layer. For example, a filter shape shift affecting a single channel might be recovered by a smart tuning of the transmitter card (e.g., a proprietary FEC adaptation). Moreover, the feedback granularity and pace that an analytics handler sends to the SDN controller to continuously perform re-optimizations may be a critical aspect, yet undiscussed. Current SDN controllers (e.g., ONOS) are not designed to handle high-frequency notifications and the SDN configurations may take even minutes to become effective on the devices. In addition, the controller may be unaware of xPonder (proprietary) transmission parameters, e.g. FEC while applying soft failure recovery [7]. In general, a mechanism aiming to segregate finer lightpath re-optimization (triggered by the card device itself) from network-wide recovery (triggered by the SDN controller) would improve the overall system scalability and efficiency.

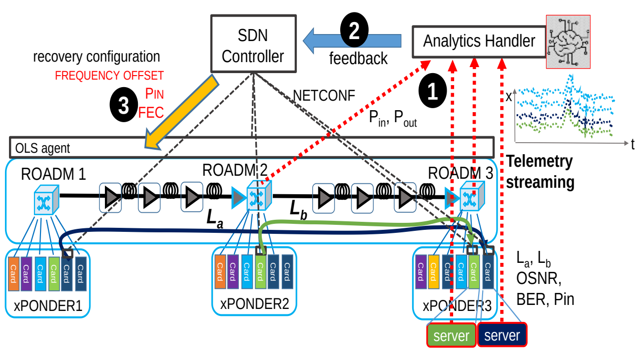

Fig. 1. Centralized telemetry service workflow.

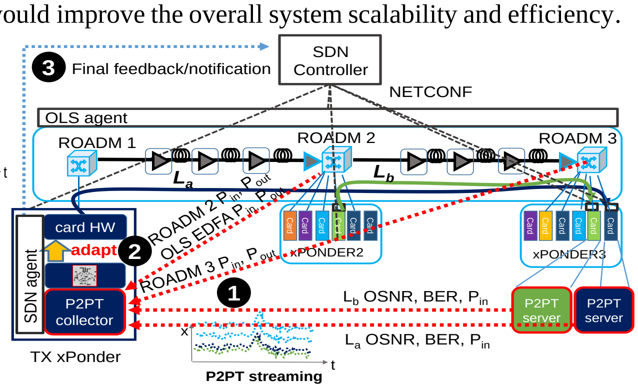

Fig.2. Proposed P2PT workflow.

3. A novel optical telemetry scheme: peer-to-peer Telemetry

The proposed Peer-to-Peer Telemetry (P2PT), in combination with standard centralized telemetry, is conceived to bring network awareness at the next-generation (pluggable) optical card, possibly equipped with lightweight storage, processing, and AI (e.g., GPU) resources. P2PT may be activated upon card configuration (i.e., at provisioning) or on-demand upon notifications during soft failures. The purpose of P2PT is twofold: 1) enable an additional hierarchical degree in the telemetry architecture, thus improving scalability; 2) enable the xPonder to take local lightpath-based decisions within the SDN controller configuration margins, based on telemetry data processed at the card itself. In both cases, as shown in Fig. 2, the SDN controller selects a set of devices and optical parameters to be monitored (e.g., as in Fig. 1), configuring telemetry subscriptions having the L_a transmitter card as telemetry stream collector. Telemetry data streamed to the transmitter (step 1) are consumed locally to identify and trigger possible soft failure recovery at the card directly (step 2), within the constraints imposed by the current SDN controller configurations. For example, fine central frequency tuning within the allocated frequency slot, proprietary FEC or constellation shaping adaptation, launch power variation are possible countermeasures that may enforced directly by the card agent and subject to subsequent monitor via P2PT. To this end, to avoid YANG extensions to SDN control models (i.e., OpenConfig, OpenROADM, TIP OOPT), the association between the stream data and the related device and parameter information is carried out by the telemetry protocol. In this paper we assume the gPRC protocol, conveying an extended *template-id* field (the device type: xPonder, amplifier, ROADM degree) and an *observation-point* defining the device id (i.e., its control plane IP address). This way the card is aware of the type of device sourcing telemetry to be consumed by local AI engines to trigger fast online soft failure recovery, without resorting to central telemetry collectors and the SDN controller loop. Notifications are assumed to acknowledge the SDN controller about the outcome of local recovery (step 3). With respect to existing solutions, P2TP extends the concept of the optical supervisory channel, introducing programmability and the flexibility to include additional telemetry sources (e.g., adjacent channels performance). This way, thanks to streaming scalability, P2PT breaks the rigidity of Optical Transport Network (OTN) hierarchy using out-of-band 10Gbps control channels nowadays available at the node level.

4. Experimental validation and results

The P2PT has been implemented and evaluated in SDN ONOS-controlled network testbed reproducing Fig. 2, including two Lumentum Greybox ROADM-20 degrees (transit ROADM2), 2 Ericsson SPO 1400 100G PM-QPSK xPonders and 2 OLS EDFA. NETCONF agents based on ConfD tool run the OpenROADM model (ROADM and EDFAs) and the OpenConfig model (xPonders). P2PT is implemented as a gRPC out-of-band streaming service available at the agents. A lightpath (193.7 THz central frequency, 37.5 GHz channel width, FEC1) is configured at the xPonders crossing the EDFAs and the Lumentum degrees, triggering P2PT subscriptions to the TX xPonder agent.

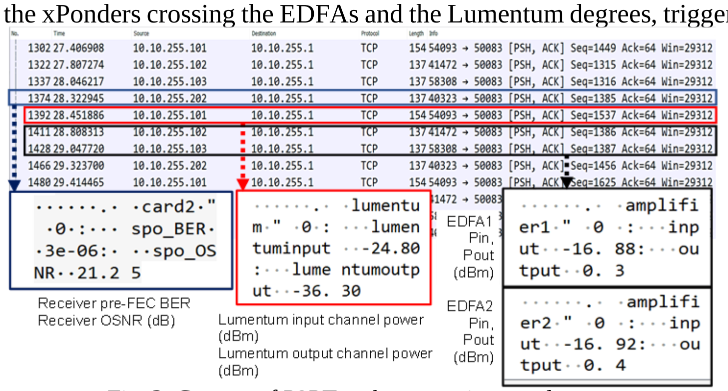

Fig. 3. Capture of P2PT at the transmitter card agent.

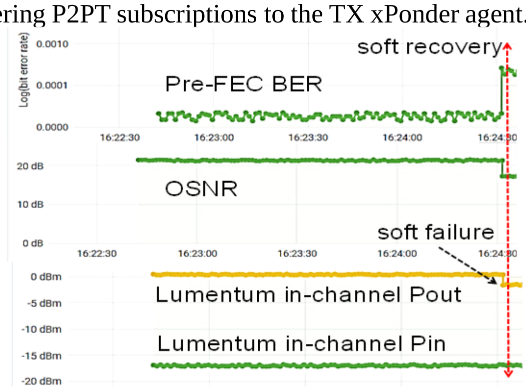

Fig. 4. P2PT monitors: soft failure experiment.

Fig. 3 shows P2PT streams captures at the TX (IP 10.10.255.1) with 4 subscriptions at 1s sampling rate (SPO receiver 202, Lumentum 101, EDFA1 102, EDFA2 103). The aggregated streaming rate is 4kbit/s average and 250kbit/s peak, including TCP overhead. This means that a 10G control plane interface is able to support up to 40k cards receiving the same amount of data, thus demonstrating P2PT scalability at the optical node. Fig. 4 shows the P2PT data at the TX. A filter shift soft failure at the Lumentum WSS induces a degradation of the in-channel output power (-1.5 dB), impacting receiver OSNR (-2dB) and BER degradation ($1.5x10^{-4}$). The card, using a Linux script, identifies the type of soft failure and, after 3 degraded samples (3s), triggers adaptation to FEC2, more robust to narrow filtering. The whole ONOS intervention would have required at least 3s more due to recomputation and NETCONF exchanges [7].

Conclusions

Telemetry architectures were discussed in the context of disaggregation, highlighting possible scalability issues and limited awareness at the controller. The proposed gRPC P2PT triggering soft failure recovery at the card showed its scalability at the node, fast reaction and best performance recovery selection without involving the controller.

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References

[1] L. Velasco et al., “Building Autonomic Optical Whitebox-based Networks”, JLT, vol. 36, n.15, pp. 3097-3104, 2018.
[2] F. Paolucci, A. Sgambelluri, F. Cugini, P. Castoldi, “Network Telemetry Streaming Services in SDN-based Disaggregated Optical Networks”, JLT, vol. 36, n.15, pp. 3142-3149, 2018.
[3] N. Sambo, F. Cugini, A. Sgambelluri, P. Castoldi, "Monitoring Plane Architecture and OAM Handler," JLT, 34 (8), pp. 1939-1945, 2016.
[4] J. Kundrat et al., “Opening up ROADMs: a filterless add/drop module for coherent-detection signals”, JOCN, vol 12, n. 6, pp. C41-C49, 2020.
[5] A. Clemn, E. Voit, “Subscription to YANG Notification for Datastore Updates”, RFC 8641, IETF, 2019.
[6] R. Vilalta et al, “Telemetry-enabled Cloud-native Transport SDN Controller for Real-time Monitoring of Optical Transponders Using gNMI”, ECOC 2020.
[7] A. Sgambelluri, A. Giorgetti, D. Scano, F. Cugini, F. Paolucci, “OpenConfig and OpenROADM Automation of Operational Modes in Disaggregated Optical Networks, IEEE Access, vol. 8, pp 190094-190107, 2020.